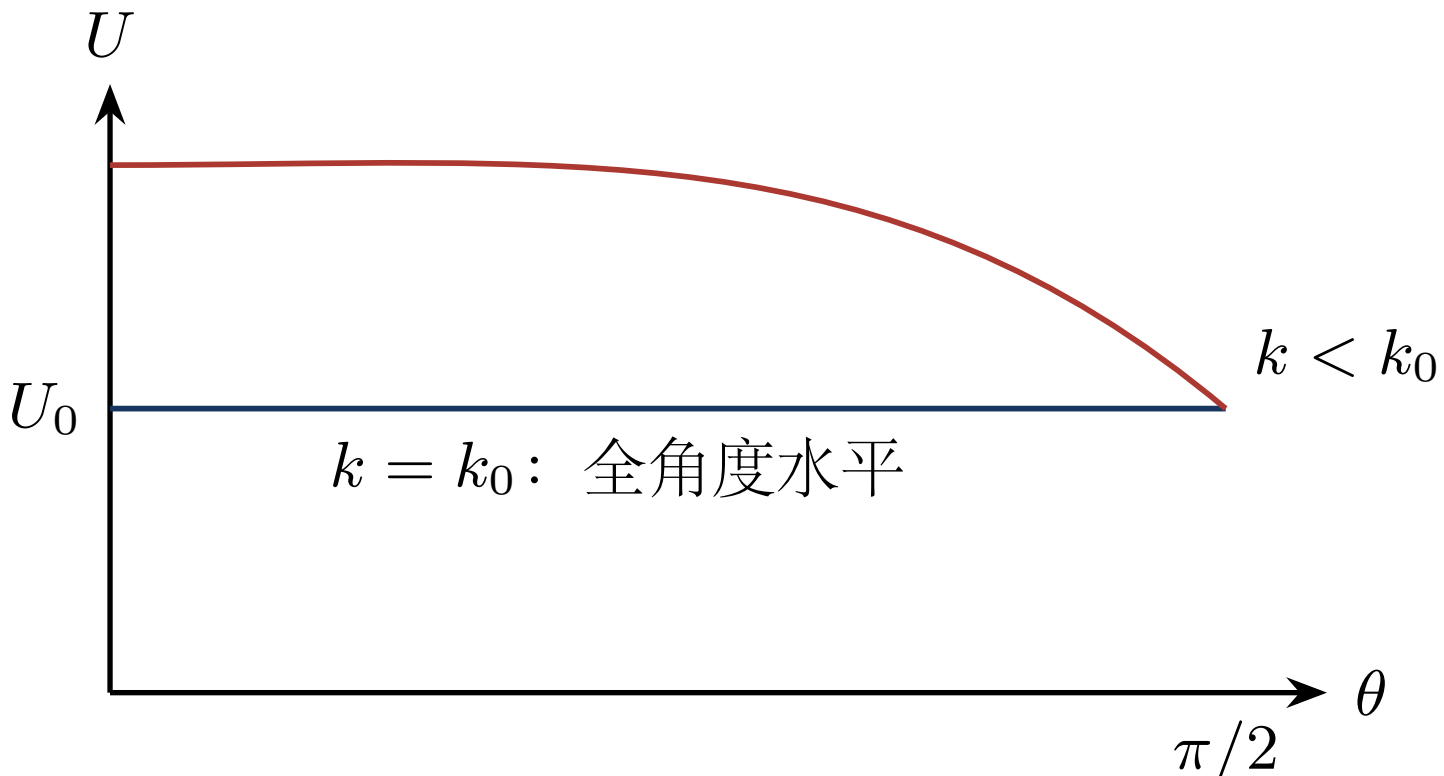

$$U = U_0 + [gL(P_1 + P_2/2) - kab]\cos\theta$$